Reinforcement Learning for Synthetic-to-Real Sensor Calibration

Technical design report and proof-of-concept plan | Radar-first, extensible to LiDAR–camera fusion

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RECOMMENDATION Use constrained RL to calibrate a low-dimensional, physically interpretable synthetic radar sensor model. Compare against Bayesian optimization, random search, and supervised system identification. Promote the PoC only if held-out real tracking performance improves.

Executive decision

This PoC uses reinforcement learning (RL) as an optimizer of simulator parameters, not as an autonomous-vehicle control policy. The agent observes discrepancies between synthetic and real radar features and changes parameters governing detection, clutter, measurement noise and temporal behavior. An inexpensive real-data calibration set may be mostly unlabeled; however, independent labeled real sequences remain necessary to substantiate tracking improvements. The technical hypothesis is that conditional, physically valid radar calibration improves real-world tracking transfer more reliably than global distribution matching or untuned domain randomization.

Decision	PoC choice	Rationale
First modality	4D radar detection/point-cloud level	Reduces simulation complexity; leverages Doppler and tracking expertise
First downstream task	Moving-object tracking	Evaluates temporal characteristics and data-association value
Optimization	Constrained contextual RL; test PPO/SAC only after simpler baselines	RL is justified only if sequential calibration beats simpler optimizers
Target real data	Small calibration set + separate held-out labeled test set	Prevents leakage and quantifies real transfer
Acceptance	Improved held-out real tracking with no degradation in safety-relevant slices	Distribution similarity is insufficient

1. Problem definition

Let s denote a physical scene (geometry, object motion, material and weather context), y the simulator output, x a real sensor observation, and θ the tunable sensor parameters. Synthetic observations are generated as $y = G(s; \theta, \varepsilon)$. The task is to choose θ, or a distribution $q(\theta \mid c)$, that improves alignment between $P_{sim}(y \mid c)$ and $P_{real}(x \mid c)$ for matched context c while retaining geometry, temporal consistency, and downstream utility. This is conditional domain matching —not a claim to recover the true data-generating mechanism.

Crucial distinction: Covariate shift means $P_{sim}(X)$ differs from $P_{real}(X)$ while $P(Y|X)$ remains invariant; real radar simulation can also exhibit conditional/label shift, missing physical phenomena, and mismatched scene priors. The PoC must therefore control scene mix and evaluate transfer, not just minimize one unconditional feature distance.

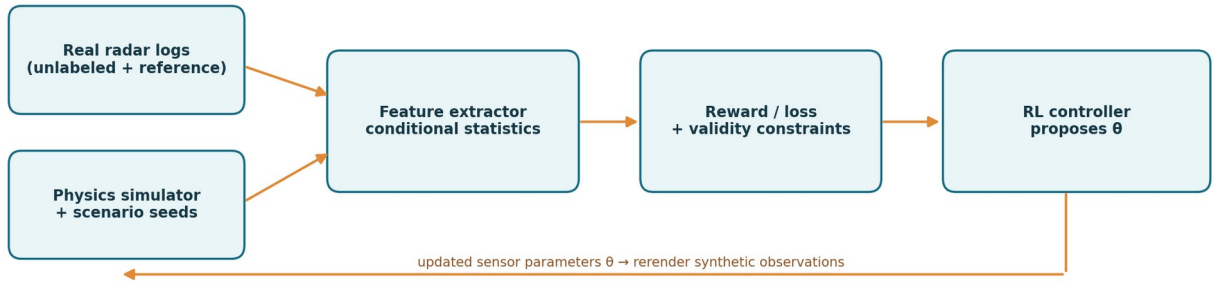
2. Scope and inputs

In scope	Out of scope
One radar model + one site/operating domain	Complete electromagnetic waveform / raw ADC radar simulation

4D radar points: range, azimuth, elevation, Doppler, intensity if available	Joint end-to-end camera–LiDAR–radar photorealistic simulation
Matched or stratified real/synthetic scenarios	Safety integrity proof or universal transfer guarantees
Offline batch optimization; no robot actuation	Online sensor alteration on production vehicles

Minimum dataset: (a) real radar sequences including sensor metadata, ego-motion and representative moving objects; (b) simulated counterpart scenarios with known object states; (c) context metadata (range, relative speed, angular sector, object type/aspect where known); (d) a held-out real test partition with independent track/kinematic reference labels for quantitative evaluation. Matched real/sim scenes are preferable but not mandatory if context strata are comparable.

3. Agent environment and architecture



The training loop is a controlled, offline calibration environment. A batch of real sensor windows and matched/context-stratified simulated windows forms one episode. After the agent proposes parameter changes, the simulator regenerates synthetic radar observations under fixed scenario seeds and the environment computes a reward. Costs are measured in simulator evaluations, not real driving episodes.

Element	Definition
State / observation	Normalized conditional feature discrepancies, previous θ, confidence intervals, prior reward, constraint violations, context embedding
Action	Bounded Δθ for physical parameters, or an index selecting a preapproved parameter update; action masking for invalid choices
Transition	Apply θ, regenerate seeded synthetic windows, estimate feature distributions and constraints
Reward	Negative conditional distribution loss minus physics penalties and evaluation cost; optional sparse downstream proxy
Termination	Budget exhausted, converged validation loss, invalid model, or maximum iterations
Safety constraint	Only physically admissible parameter configurations and non-destructive offline evaluation

4. Radar parameterization: start small

Parameter θ	Initial range (illustrative)	Observable effect
Range noise σ_r	0.03–0.35 m	Range residual spread
Azimuth noise σ_az	0.2–3.0 degrees	Angular uncertainty / association errors
Doppler noise σ_v	0.05–0.8 m/s	Velocity-estimation spread

Detection probability parameters	Logistic curve vs range / RCS proxy	Return counts by distance, aspect and speed
Clutter rate λ_c	0.0–8 candidate returns/frame/sector	False returns and spatial occupancy
Missed-detection burst length	1–8 frames	Track fragmentation and temporal dropout
Time correlation coefficient ρ	0.0–0.95	Temporally correlated measurement disturbances

Ranges are proposed engineering search bounds only; replace them with the radar manufacturer's data sheet, empirical calibration, and physically plausible limits. Prefer simulator-native parameters; allow a residual stochastic postprocessor only where the base simulator cannot represent documented phenomena. Keep 6–10 adjustable dimensions initially to avoid identifiability and sample-efficiency problems.

5. Feature design and domain discrepancy

Do not compare raw global point clouds directly: differences in scene composition can dominate the metric. Build context-conditioned summaries, with real and synthetic windows matched by ego-speed, range, azimuth sector, object motion, and where possible geometry/material. Recommended features: detection counts per range bin; Doppler conditional on target kinematics; range and angle residuals against surveyed targets; clutter spatial statistics; run lengths of missed detections; tracklet length; inter-frame persistence; and point-density covariance across modalities when available.

For each context bin c, define d_c as a weighted mix of (i) sliced Wasserstein distance for continuous features; (ii) maximum mean discrepancy (MMD) for feature embeddings; (iii) histogram divergence for counts; and (iv) differences in temporal autocorrelation. Use bootstrap confidence intervals over independent sequences/sites rather than treating frames as independent. Use normalized, predeclared weights to prevent rewards from being dominated by a single high-variance statistic.

Illustrative objective: $L_{domain}(\theta) = \Sigma_c w_c [\alpha SWD(F_{real,c}, F_{sim,c}(\theta)) + \beta MMD^2(F_{real,c}, F_{sim,c}(\theta)) + \gamma D_{temporal,c}(\theta)]$. Optimize the conditional mismatch only where both domains have adequate support. Report unsupported contexts separately rather than extrapolate a score.

6. Reward specification

At optimization iteration k, define $r_k = -\Delta L_{domain} - \lambda_p C_{physics} - \lambda_t C_{temporal} - \lambda_e C_{eval} - \lambda_o C_{overfit}$. Using improvement ΔL rather than absolute loss can stabilize learning when scenario difficulty varies. $C_{physics}$ penalizes violations such as inconsistent range/Doppler kinematics; $C_{temporal}$ penalizes unrealistic frame-to-frame variations; C_{eval} is compute cost; $C_{overfit}$ is a gap between calibration and internal validation performance.

Task-aware reward: after a cheap initial search, periodically add an independently computed tracking proxy or validation score: $r_{task} = \eta [M_{track}(\theta) - M_{track}(baseline)]$. Do not tune parameters directly against the final held-out test set. If labeled validation data are scarce, retain a task-blind calibration loop and use task-level evaluation only to select among candidate checkpoints on a separate validation partition.

Hard feasibility constraints (e.g., monotonic detection-probability envelope where physically justified, bounded noise, correct sign/units of Doppler, temporal plausibility) should be enforced by parameter transforms or simulator constraints rather than relying solely on a penalty.

7. Is reinforcement learning the right optimizer?

This is an important risk: static calibration of a low-dimensional black-box simulator is often better addressed by Bayesian optimization, covariance-matrix adaptation (CMA-ES), or derivative-free search than by RL. RL becomes defensible if the company needs sequential, adaptive calibration policies across many radars, contexts, simulator versions, or nonstationary domains. The PoC should test RL as a hypothesis, not assume it is superior.

Method	Role	Evaluation
Hand-tuned calibration	Operational baseline	Expert time and transfer quality
Random search / Latin hypercube	Simple search baseline	Best reward per 100 simulation calls
Bayesian optimization	Strong low-dimensional baseline	Sample efficiency and wall-clock cost
CMA-ES	Strong derivative-free baseline	Robustness to nonconvexity
Contextual PPO or SAC	Proposed adaptive RL optimizer	Generalization to new contexts and optimization budget
Ablation: no temporal constraints	Tests reward design	Whether matching snapshots harms tracking

Implementation choice: first model the problem as a contextual bandit / finite-horizon MDP with 5–15 updates per episode, not thousands of low-value RL steps. Use PPO for bounded continuous $\Delta\theta$ and stable on-policy updates, or SAC if simulations are expensive and replay efficiency matters. Train across many simulator/domain variants; otherwise the policy may merely memorize one calibration problem.

8. Training protocol and prevention of leakage

- Partition by site, date, scene and sensor configuration—not random frame splitting. Calibration logs, internal validation logs, and final test logs must remain separate.
- Fix a suite of scenario seeds for paired optimizer comparisons; use additional seeds for robustness evaluation.
- Fit feature scalers and weights using calibration data only; never inspect final-test labels during RL optimization.
- Apply the learned θ to synthetic training sequences, then train the same downstream tracker/model under fixed compute, architecture and real-label budgets.
- Compare on held-out real sequences including range, speed, occlusion, clutter and unseen-scene slices; report confidence intervals over independent sequences.
- Retest calibration on a second radar/setting to establish portability; otherwise label the result sensor-specific.

9. PoC experimental matrix

Arm	Synthetic calibration	Training real labels	Purpose
A	Uncalibrated simulator	Fixed budget N	Baseline reality gap
B	Wide domain randomization	Same N	Robustness baseline
C	Hand-calibrated simulator	Same N	Human expert baseline
D	Bayesian-optimized parameters	Same N	Strong optimizer control
E	RL-calibrated parameters	Same N	Primary hypothesis
F	RL without temporal penalty	Same N	Temporal ablation

Where a tracker is pretrained synthetically and fine-tuned with real data, repeat each arm at N = 0, small N, and moderate N labeled real trajectories, holding all other factors fixed. Prefer 3–5 model training seeds where compute permits. The final real test set is never used for reward optimization.

10. Success metrics and gates

Layer	Primary metric	Proposed PoC gate (illustrative)
Sensor fidelity	Conditional SWD/MMD plus temporal discrepancy	≥20% reduction vs uncalibrated; no major slice regression
Real-world tracking	HOTA; velocity MAE; track fragmentation	Measurable positive lift vs best non-RL baseline with uncertainty interval

Data efficiency	Real-label learning curves	Match baseline quality with materially fewer labeled real samples
Physics validity	Constraint-violation count	Zero hard-constraint violations
Compute efficiency	Sim calls / wall-clock / GPU-hours	RL justified vs Bayesian optimization on amortized workload
Transfer	Repeat on held-out sensor/site context	Benefit persists; otherwise scope claims narrowly

Go/no-go: A lower domain discrepancy is necessary but not sufficient. Advance only if real held-out tracking improves over both uncalibrated and well-calibrated non-RL alternatives, and the gain justifies the adaptation cost. If RL ties Bayesian optimization, deliver the simpler optimizer in the product.

11. Eight-to-twelve-week execution plan

Week	Work package	Milestone
1–2	Select radar, recording schema, scene strata and held-out splits	Frozen experiment charter
3–4	Sensor statistics + synthetic pipeline + sanity tests	Trustworthy objective computation
5–6	Random search / BO / CMA-ES; physics constraints	Strong baseline calibration
7–8	Contextual RL environment and policy; ablations	RL candidates and compute comparison
9–10	Train/evaluate fixed tracking architecture on real holdout	Task-level evidence
11–12	Replication on second condition; packaging/demo	Go/no-go memo and SDK prototype

Team: one senior radar/sensor-physics engineer, one ML/RL engineer, one perception/tracking engineer, and shared data/DevOps support. Major dependencies: usable simulator hooks; real logs with metadata; a small survey/calibration set for conditional sensor metrics; and independent real tracking evaluation sequences.

12. Risks, mitigations, and fallback paths

Risk	Consequence	Mitigation
Scene-mix confounding	Optimizer calibrates scenario frequencies, not sensor behavior	Conditional stratification and paired scenarios
Non-identifiable θ	Different parameter sets yield similar statistics	Reduce dimension, priors, controlled targets, uncertainty bounds
Reward hacking	Pretty statistics, worse tracking	Task holdout; physics checks; temporal tests
Simulator missing a mechanism	Irreducible residual gap	Residual error model + explicit unsupported-domain reporting
RL sample inefficiency	Higher cost than BO	Benchmark against BO/CMA-ES and stop early
Distribution shift to new sensor/site	Loss of validity	Context-conditioned adaptation and out-of-support detection
Missing real evaluation labels	Cannot substantiate product performance	Budget targeted reference data; do not claim proven transfer

13. Commercialization hypothesis

Initial product: simulator-independent Sensor Reality Calibration SDK operating on radar detection outputs and simulator parameter APIs. It should export calibrated sensor profiles, discrepancy reports, and reproducible synthetic regeneration configs. An enterprise service can initially help customers instrument their radar models, but each engagement should produce reusable software rather than permanent bespoke consulting.

The first buyer is an organization already spending materially on synthetic radar data and training radar–camera–LiDAR perception models. The purchase trigger is improved real-world tracking or a lower real-label requirement, not abstract “distribution alignment.” Start with one customer-funded comparison against its existing synthetic-data pipeline. Longer term, extend to joint multimodal conditional alignment rather than independently beautifying each sensor stream.

14. Relevant research and technical references

Trinh et al. (2026), A Physics-Informed Digital Twin Framework for Calibrated Sim-to-Real FMCW Radar Occupancy Estimation — https://openjournals.uwaterloo.ca/index.php/vsl/article/view/7141

Bejerano et al. (2026), Sim2Radar: Toward Bridging the Radar Sim-to-Real Gap with VLM-Guided Scene Reconstruction — https://arxiv.org/abs/2602.13314

TUM (2025), Modeling and Simulation of Automotive FMCW RADAR Sensor for Environmental Perception — https://portal.fis.tum.de/en/publications/modeling-and-simulation-of-automotive-fmcw-radar-sensor-for-envir/

Muratore et al. (2022), Robot Learning From Randomized Simulations: A Review — https://pmc.ncbi.nlm.nih.gov/articles/PMC9038844/

Muratore et al. (2021), Assessing Transferability From Simulation to Reality for Reinforcement Learning — https://www.dfki.de/en/web/research/projects-and-publications/publication/13581

Chen et al. (2022), Understanding Domain Randomization for Sim-to-Real Transfer, ICLR — https://mlanthology.org/iclr/2022/chen2022iclr-understanding-a/

Evidence note: The paper references establish the feasibility of calibrated synthetic sensor data and the risks of simulator overfitting. They do not demonstrate that the specific RL approach proposed here outperforms derivative-free calibration or generalizes across arbitrary sensor families. All tuning ranges, thresholds, timelines, and pricing implications are design assumptions to be tested.